

Weekly Progress Report

AgAI Pesticide Residue & Leaf Classification Project

Week 1: 2/23/2026

Overview

This week I focused on onboarding into the AgAI project, understanding the technical progression from last year's work, and clarifying the modeling objectives for this year.

Meetings and Technical Alignment

- Met with Giulio Diracca and Melissa Deinys at the NSTC lab in Research I.
 - Reviewed the evolution of the project from last year's single-label leaf classification model (clean / canker / dirt / etc.) to this year's expanded objectives.
 - Confirmed that this year's goals include:
 - Multi-label classification (e.g., canker and dirt simultaneously).
 - Predicting pesticide residue concentration across four discrete classes (400, 600, 800, 1000 PPM).
 - Discussed dataset structure and constraints, including ICP-based ground truth validation.
- Connected with William Bu to better understand last year's leaf classification pipeline architecture.
 - Learned that the prior system used RF-DETR for detection, FastViT as a lightweight backbone, and a classification head.
 - Discussed scheduling a Teams call with Anthony Bilic to brainstorm and formalize this year's pipeline direction.
- Reached out to Anthony Bilic to learn more about last year's implementation details and align on this year's modeling approach.

Dataset Assessment

- Current dataset availability:
 - ~75 leaves per plant
 - 4 plants per class
 - ~300 leaves per class (approximately 1,200 total leaves)
- Only ~5 leaves per class undergo ICP (Inductively Coupled Plasma) validation due to the destructive and expensive measurement process.
- Remaining leaves are assigned estimated PPM values.

Key considerations identified:

- 300 leaves per class is near the lower bound for statistically meaningful training in a lightweight transformer-based model.
- Limited ICP-validated samples introduce label reliability concerns.
- Plant-level diversity is limited (4 plants per class), increasing risk of overfitting if not split carefully.
- The biological objective is ultimately threshold-based (determining whether pesticide levels fall below protective concentration), which may influence final model formulation.

Technical Considerations Identified

- 300 leaves per class is near the lower bound for statistically meaningful training in a lightweight transformer-based model.
- Label reliability is a concern due to limited ICP-validated ground truth.
- Potential risk of overfitting due to limited biological diversity (only 4 plants per class).
- The ultimate biological objective is determining whether pesticide levels fall below protective threshold (i.e., whether reapplication is needed), suggesting a potential binary decision framing alongside discrete PPM classification.

Environment and Research Integration

- Established workspace in the IAI lab and began integrating with the broader research team.
- Plan to maintain consistent lab presence on Tuesdays and Thursdays from 1–6 PM to fulfill 10-hour minimum weekly research commitment and support collaboration.
 - This upcoming Thursday (3/5) I will not be working during my usual window.

Next Steps

- Review last year's classification architecture (RF-DETR + FastViT backbone + classification head).
- Research multi-label classification approaches to support simultaneous condition predictions.
- Explore modeling strategies to handle limited ICP-validated samples (e.g., weighted loss, weak supervision, or threshold-based binary classification).
- Evaluate whether a direct 4-class PPM model or a threshold-based model better aligns with the biological objective.